1)
$$\begin{vmatrix} 3 & 6 & 1 & 2 \\ 0 & 6 & 3 & 5 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 3 \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} 1 \\ -1 \\ 9 \\ 3 \end{vmatrix}$$

$3x_4 = 3$ $\boxed{x_4 = 1}$

$4x_3 + x_4 = 9$ $4x_3 + 1 = 9 \rightarrow \boxed{x_3 = 2}$

$6x_2 + 3x_3 + 5x_4 = -1 \rightarrow 6x_2 + 6 + 5 = \boxed{x_2 = -2}$

$3x_1 + 6x_2 + x_3 + 2x_4 = 1$

$\rightarrow 3x_1 = 9$ $\boxed{x_1 = 3}$

2)

$$\begin{vmatrix} 0 & 1 \\ 1 & 1 \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = \begin{vmatrix} 1 \\ 2 \end{vmatrix}$$ $\rightarrow$ first equ has a coeffcient of zero for variable x_1. We need to interchange the equation.

$$-1 \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} x_2 \\ x_1 \end{vmatrix} = \begin{vmatrix} 2 \\ 1 \end{vmatrix} \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_2 \\ x_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$x_2 = 1$ $x_1 = 1$

③ $2z^5 - 13z^4 + 26z^3 - 32z^2 + 28z - 10$

upper $\rightarrow$ $P_2 = 1 + |a_n|^{-1} \cdot \max\limits_{0 \le i \le n-1} |a_i|$

$1 + \frac{1}{2} \cdot 32 = \boxed{17 = P_2}$

lower $\rightarrow$ $P_1 = \quad S(x) = x^n f(x^{-1})$

$= -10x^5 + 28x^4 - 32x^3 + 26x^2 - 13x + 2$

$\rightarrow 1 + \left(\frac{1}{10}\right) \cdot 32 = \frac{42}{10} = P_1^{-1}$

$\boxed{\frac{5}{21} = P_1}$

$\boxed{\frac{5}{21} < z < 17}$

④

x	2	-1	1	0
$f(x)$	-2	-8	2	2

0 $\quad L_0(x) = \frac{(x+1)(x-1)x}{(3)\cdot(1)\cdot 2} = \frac{(x+1)(x-1)\cdot x}{+6}$

1 $\quad L_2(x) = \frac{(x-2)(x+1)\cdot x}{(-1)(2)\cdot 1} = \frac{(x-2)(x+1)x}{-2}$

1 $\quad L_1(x) = \frac{(x-2)(x+1)\cdot x}{(-3)(-2)\cdot -1} = \frac{(x-2)(x-1)\,x}{-6}$

3 $\quad L_3(x) = \frac{(x-2)(x+1)(x-1)}{(-2)(+1)(-1)} = \frac{(x-2)(x+1)(x-1)}{2}$

$P_3(x) = -2L_0(x) - 8L_1(x) + 2L_2(x) + 2L_3(x)$

$P(x) = x^3 - 5x^2 + 4x + 1$